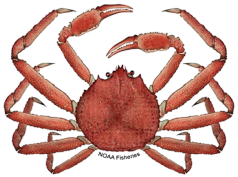
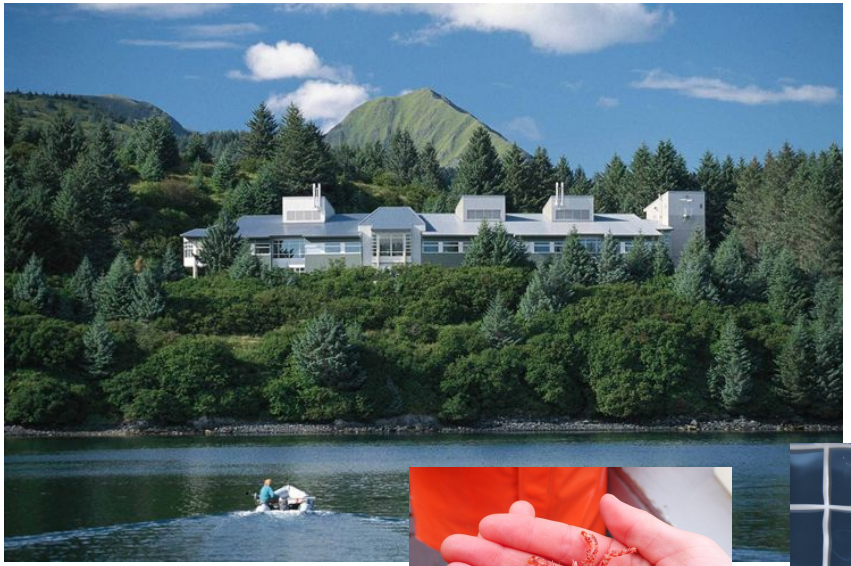
Mechanisms of pH stress resilience in Alaskan crabs



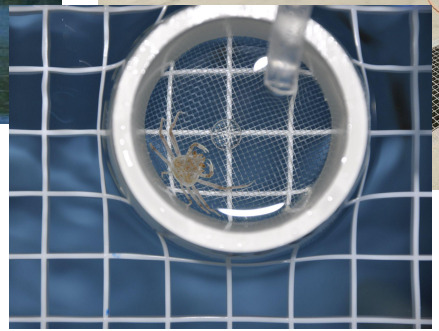
Annual Meeting, National Shellfish Association
March 2026, Portland, OR

Laura Spencer, NOAA Affiliate at Univ. of Washington
Jennifer Gardner, NOAA AFSC Kodiak
Ingrid Spies, NOAA AFSC Seattle
Steven Roberts, University of Washington SAFS
Chris Long, NOAA AFSC Kodiak

Integrating molecular tools with experimental pH stress response studies









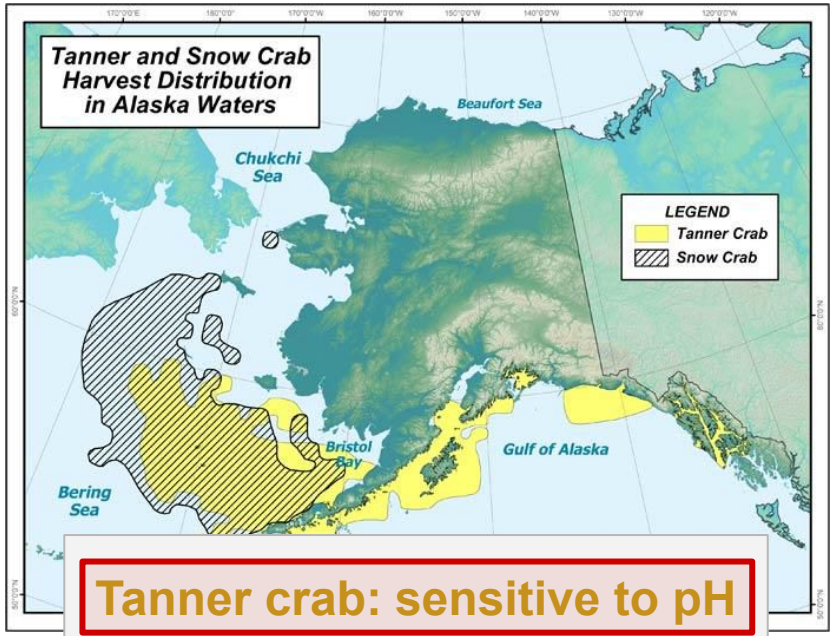
NOAA Kodiak
Research Station

Tanner & snow crabs, *Chionoecetes spp.*





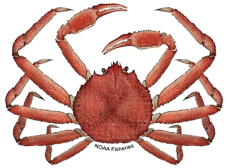




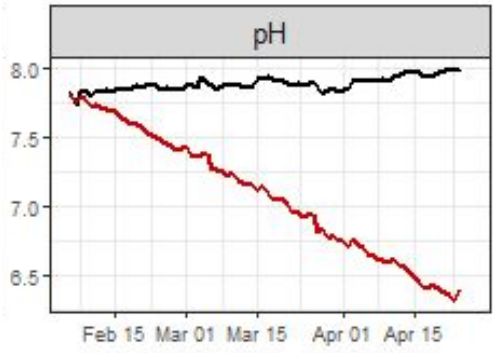
Tanner crab: sensitive to pH

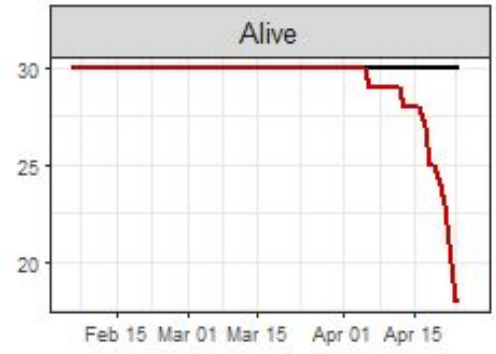
Snow crab: tolerant of pH

FG



Experiment! What is distinct about Tanners that survive low pH?



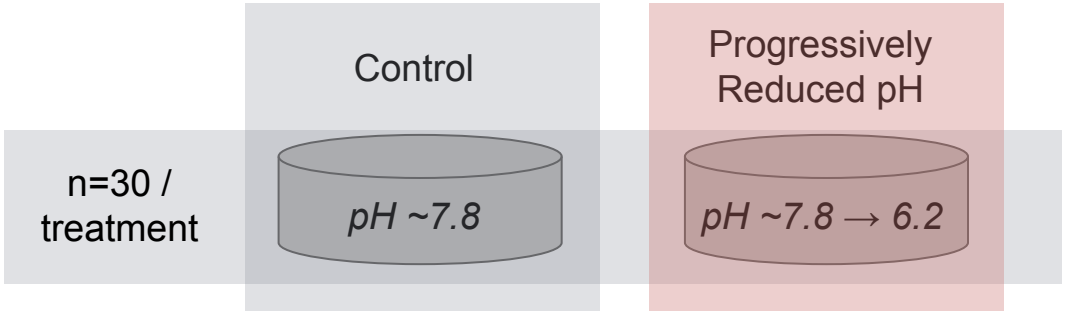


Laboratory Experiment

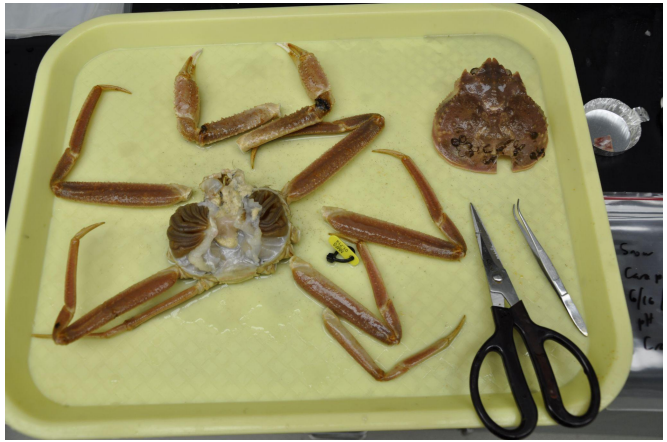
- Wild adult mature males from Eastern Bering Sea
- pH reduced over ~2 month until ~50% mortality

Data:

1. Survival, size
2. Molecular data



Experimentally derived "Genome-to-Phenome" dataset for low pH survivors, Control group

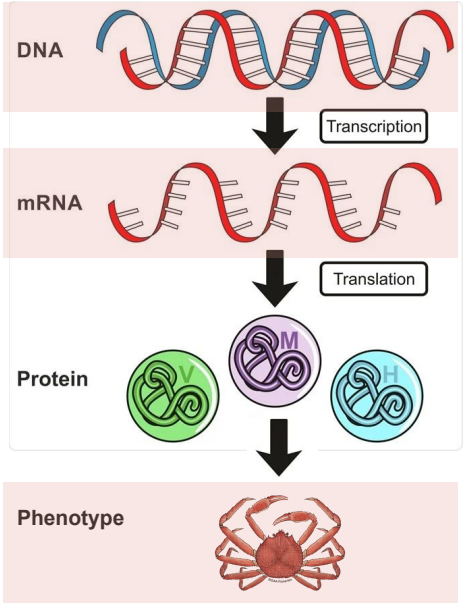


n=17 Control
n=14 low pH Survivors

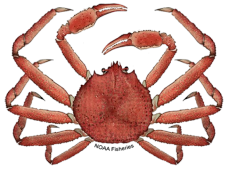
Genetic data (lcWGS)
- Whole-genomes resequenced ~2x

Expression data (RNAseq)
- Gill
- Hepatopancreas
- Muscle

Phenotypes
- Survival, Size

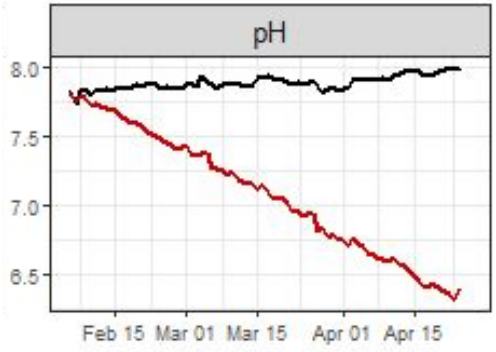


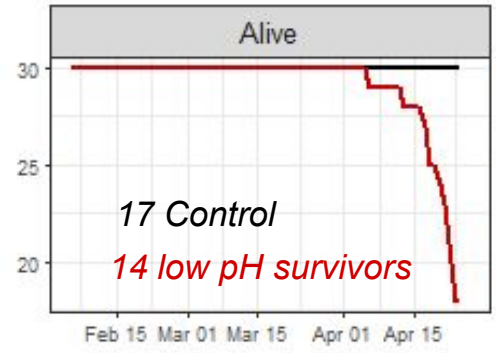
Adapted from: udaix/Shutterstock.com

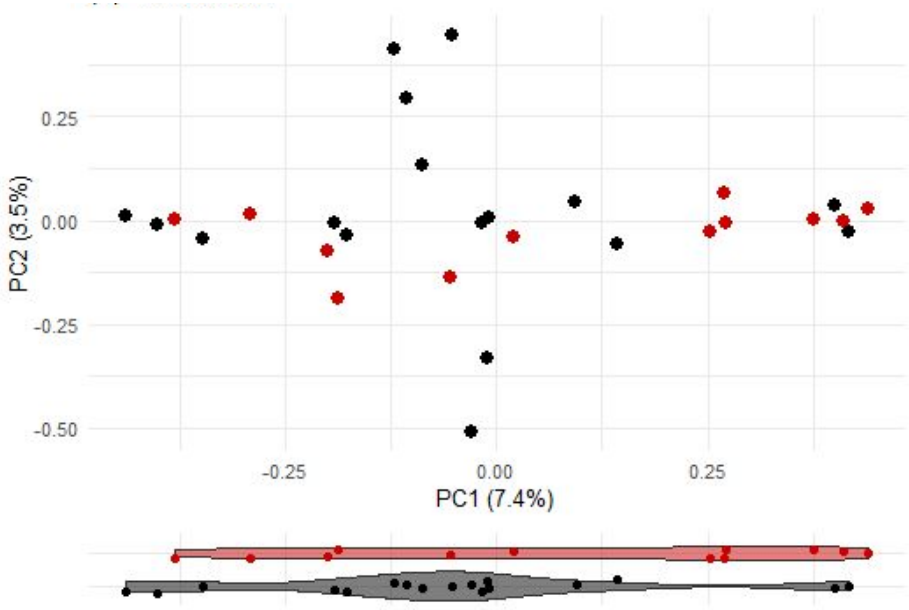


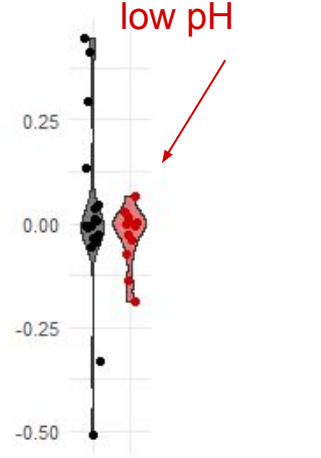
Less genetic variability in low pH survivors

PCA, genetic data (~350k SNPs)

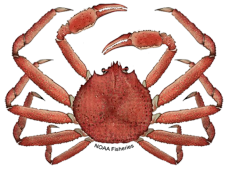






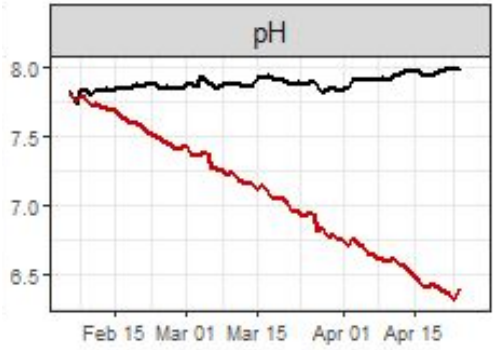


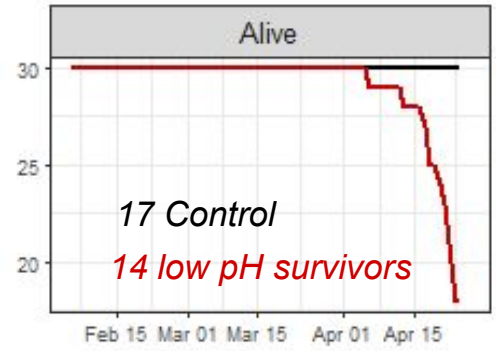
Less variability in low pH

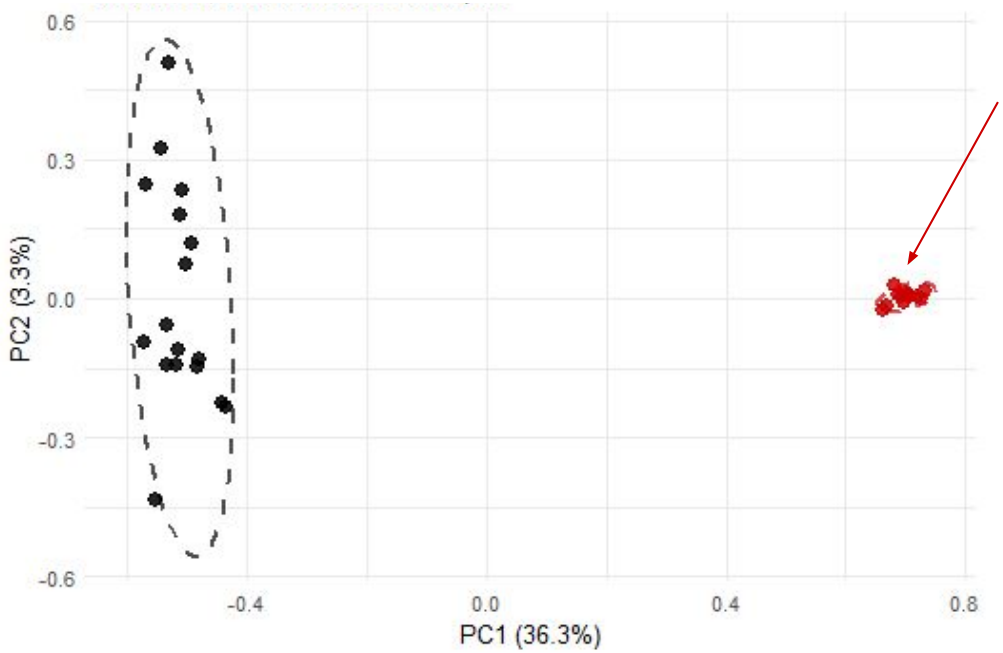


Less genetic variability in low pH survivors

PCA, SNPs with Fst > 0.1 (n=1,900)

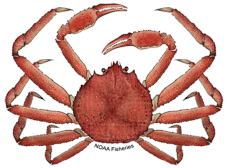






Way less variability in low pH

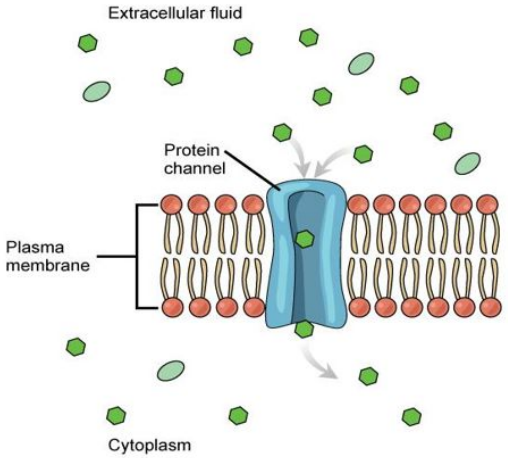
Fewer heterozygotes (individuals with multiple alleles)

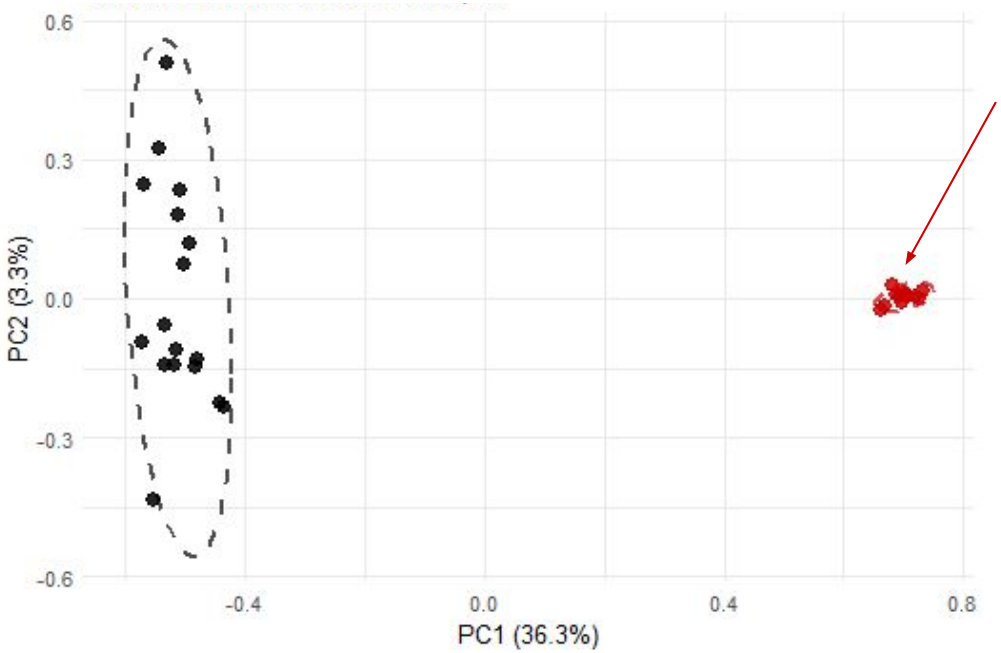


Less genetic variability in low pH survivors

PCA, SNPs with Fst > 0.1 (n=1,900)

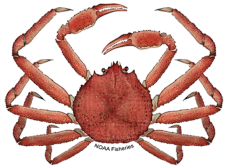
Genes containing these sites enriched for **“mono-atomic ion transport”**





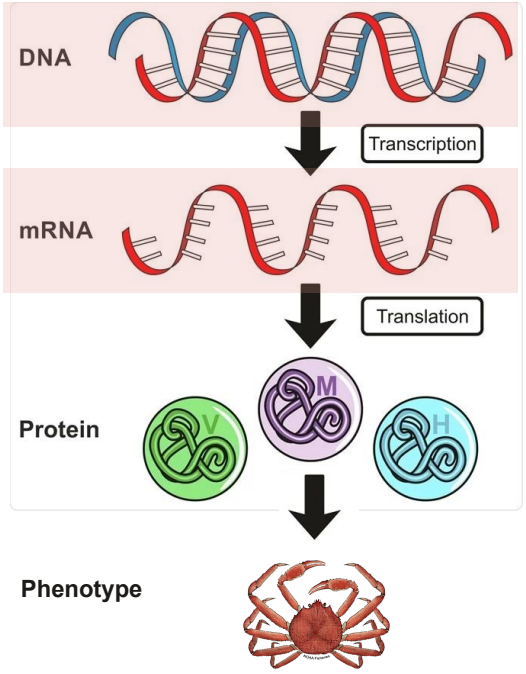
Way less variability in low pH

Fewer heterozygotes (individuals with multiple alleles)

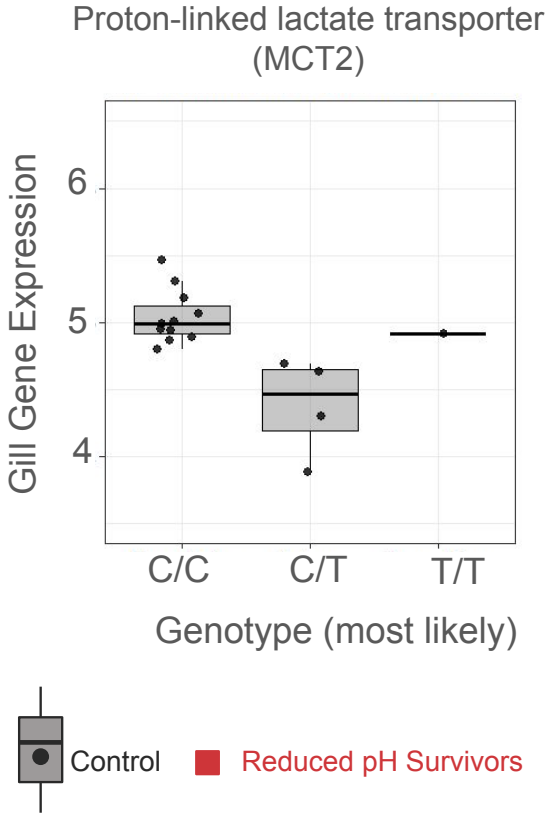


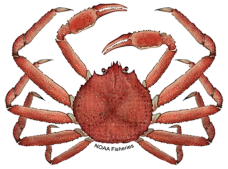
Some genetic differences may influence gene expression

(cis-eQTLs, SNPs within the same gene)

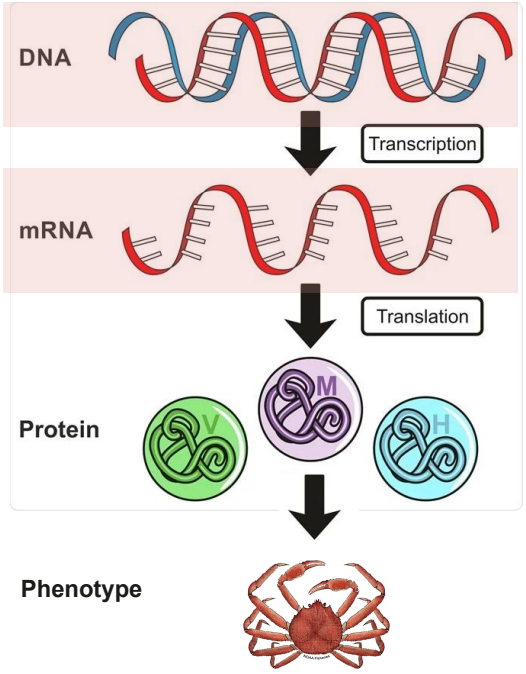


Adapted from: udaix/Shutterstock.com

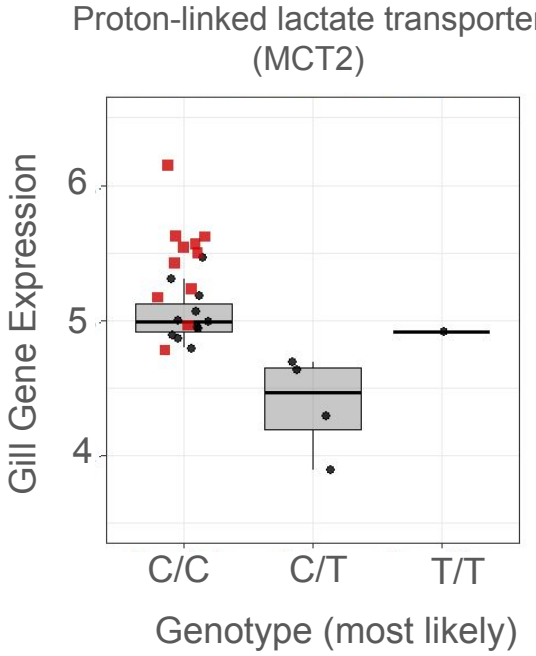


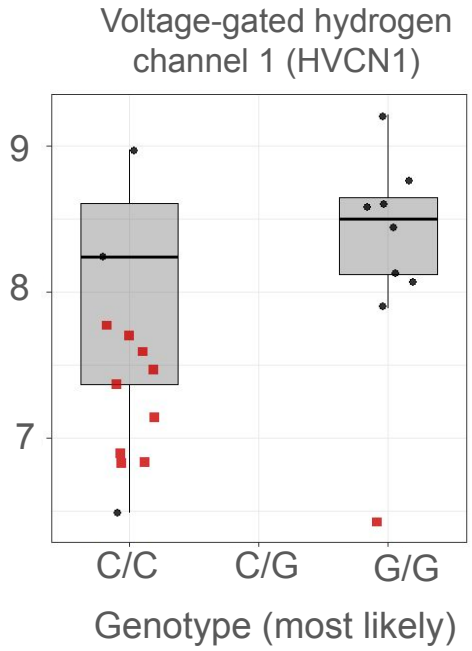


Some genetic differences may influence gene expression
(cis-eQTLs, SNPs within the same gene)



Adapted from: udaix/Shutterstock.com







Also… *trans-eQTLs*

What is distinct about Tanners that survive in OA?

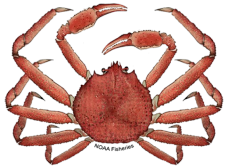


1. **Low pH survivors have less genetic variation** (more homozygotes) where there are genetic differences

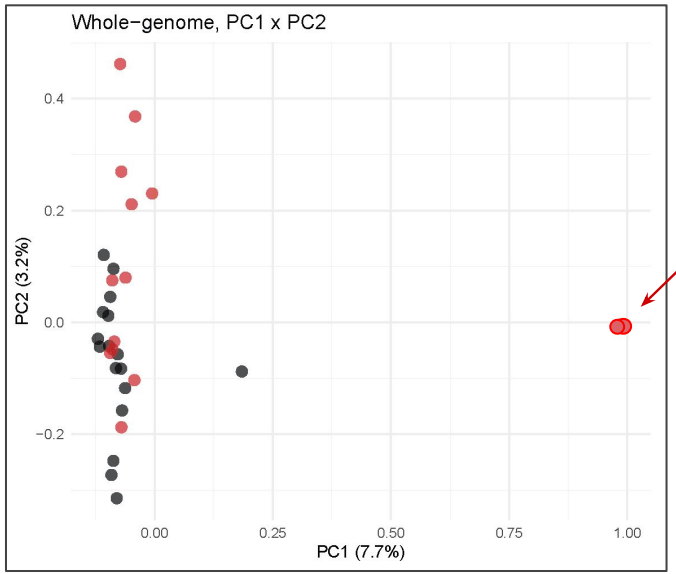
   But, very small subset of SNPs.

2. **Effective ion transport** in gill, hepatopancreas likely important for low pH tolerance.

One more thing …



Genetic outliers, who are they?

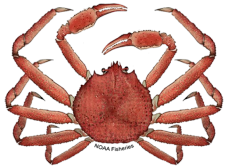


????

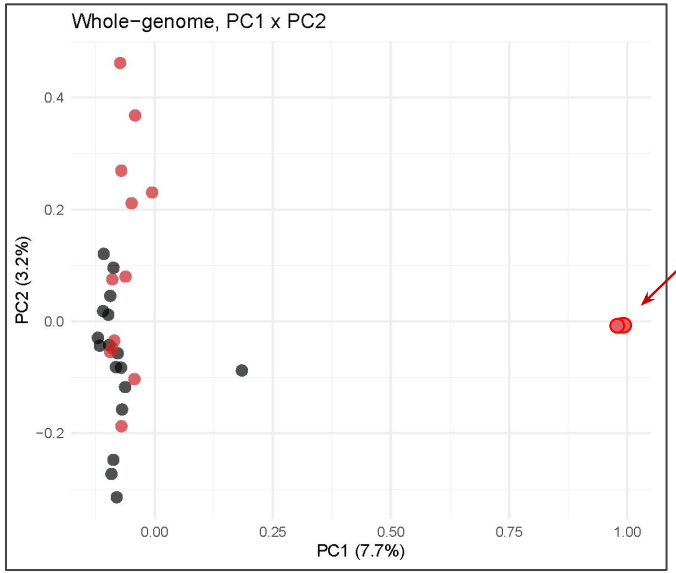
OA Treatment

Ambient

Reduced

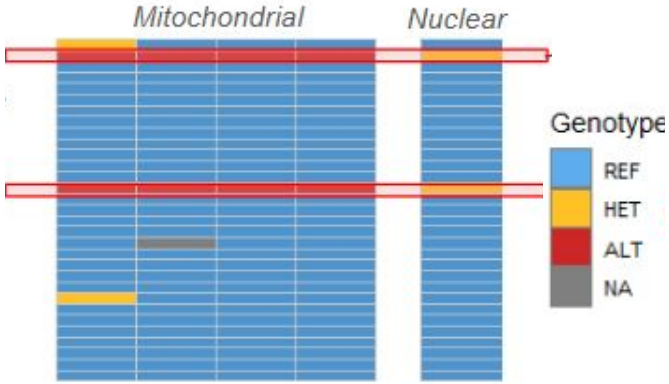


Genetic outliers, who are they?

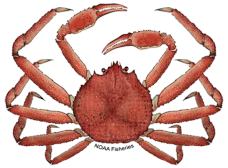


Snow-Tanner hybrids!

Two of 14 low pH survivors



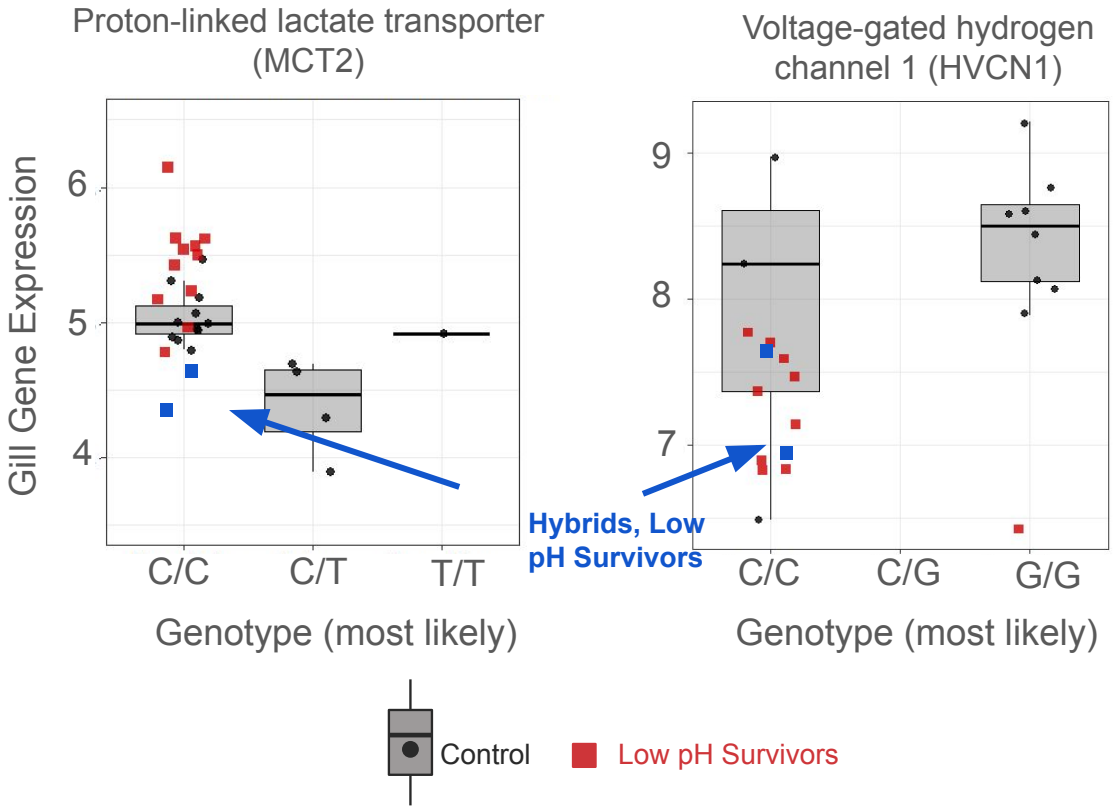
SNP that differ among tanner crab (REF) and snow crab (ALT)

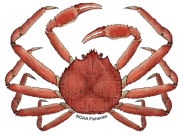


Are low pH survivors more genetically similar to hybrids?

Are hybrids more low pH tolerant?

Can snow crab transfer low pH tolerance to Tanners by interbreeding?





Limitations, Next Steps

1. Genotype likelihood data is noisy.
    a. Strict filtering and lcWGS-appropriate programs (e.g. ANGSD)
    b. Running secondary analyses with SNPs from RNASeq data.
2. Low sample sizes, limited power.
    a. Using multiple statistics (Fst, ΔAF, LRT_{GWAS}).
    b. Also identifying genome-wide SNP-DEG pairs to catalog eQTLs that could be important for adaptation (but not necessarily captured here)
3. Only two hybrids limits analyses.
    a. Comparing “snow crab allele scores” among treatments
    b. Future study needed to directly compare snow, Tanners, and their hybrids.

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